

SR Game Breakdown

Introduction

This document tries to list and sort all the key elements of the game to provide a basic mapping of the game and facilitate its expansion through:

- Additional content release
- Minor changes or improvements
- New features

Overview

SR is a level based hypercasual runner

Vertical display, the character “runs” on a 3D path, camera behind him. The main vector goes from the bottom left of the screen to the middle of the screen (approx.); the character position relatively to the player is stable, it’s mainly the path that is moving.

The player main action is to slide on the bottom of the screen; going from left to right will spread the right leg from middle (no spread) to extreme right (extreme spread) and create a mirror action from left leg. The slide is continuous, but the game uses mainly 3 positions: no spread, mid-spread, full-spread

In game, the main goal of the player is to reach the end of the level. To do so, they must not “die”. Dying occurs in 3 situations: having rollers spread corresponding to blocking-obstacles, having rollers spread not corresponding to path-obstacles, (rarer) having a too low speed for jump-obstacles. Please note that jump obstacles can also be path-obstacles.

In game, the second goal of the player is to collect soft-currency, represented by pink gems

In the Meta, the player can decide to have the following additional goals: have more characters (quality or quantity), have more rollers (quality or quantity). They could also decide to have as much SC as possible, but it’s a weaker goal, as SC is more a mean than a goal.

SR monetizes through advertisement:

- Banners
- Interstitials
- Rewarded videos
- Pay-to-remove-ads (1-time purchase or VIP subscription)

In game

Levels structure

Levels are structured into chapters of 4 levels

The first two are regular levels

The third is quite regular, but ends with giving a chest key to the player

The fourth is special, and is called "Gem Rush":

- It contains more pink gems (soft-currency) and has a gem counter



- It is easier, as most obstacles have been replaced by gems
- It ends with a jump, determining a final multiplier (x1.1 to x2, +0.1 per step)

Chapters structure: new elements (40 levels)

With this level structure, 40 levels represent 10 chapters

Each Chapter is supposed to be linked to a new Element:

- Element is "unlocked" at the end of 4th level of previous Chap
- Next unlock (end of new Chap) is indicated
- The element is used immediately, in the first level of the new Chap

Unlocks per Chap are:

- Chap1: basics
- Chap2: turn
- Chap3: boost
 - End: chestgame1
- Chap4: slopes
- Chap5: ramp
- Chap6: loop
 - End: chestgame2
- Chap7: sloping corner
- Chap8+: *nothing, except chestgame every 3 chapter*

Obstacles and spread positions

We will consider that there are 3 families of obstacles:

- Real obstacles: direct, with immediate death
- Speed obstacles: direct, without immediate death
- Global obstacles: indirect

For Real and Speed obstacles, there is a deep link between obstacles and spread positions

About spread positions

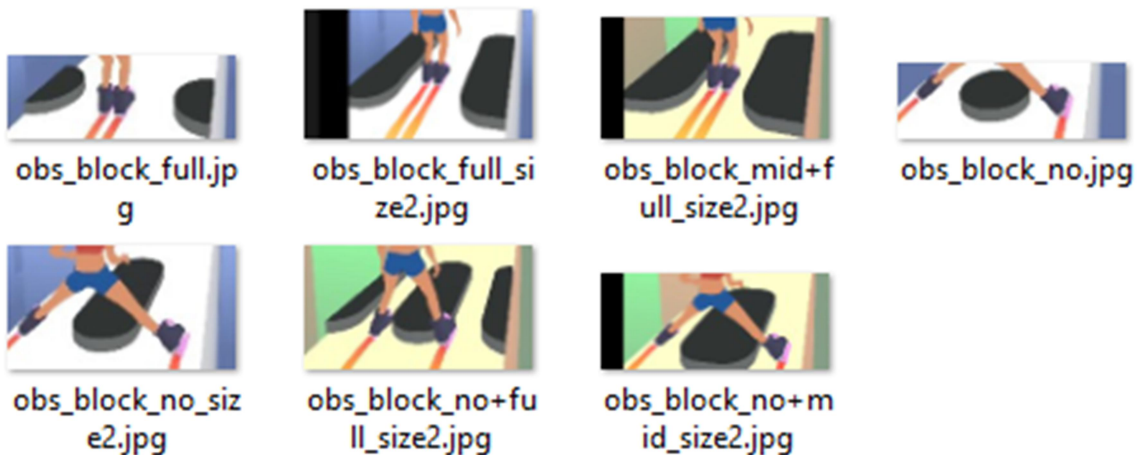
There is infinity of spread positions, but the game is based on 3 key positions, with a false 4th one

These key positions are:

- No spread (no) when the finger is totally on the left
- Medium spread (mid) when the finger is approx. in the middle of the screen
- Full spread (full) when the finger is totally on the right
- “Almost full spread” (full-) can be found in path obstacles, but such obstacles can also be passed in full spread

Blocking-obstacle (Real)

Blocking obstacles are defined by the positions they forbid, and secondarily by their size (how long they last on screen). Some forbid 1 key position, others 2



Path-obstacles (Real)

Path obstacle are defined by the positions they allow, and if they are progressive or not

Progressive path obstacles ask the player to progressively switch from a key position to another, and usually to progressively go back to initial position



Speed-obstacles (Speed) and jump-obstacles (Real)

Speed obstacles are obstacle that doesn't affect immediately win/lose, but affect speed (positively or negatively)

The player is incited to ride over positive speed obstacles and avoid negative speed obstacles

Speed obstacles can be shaped in any shape of the path obstacles

Speed is here only used in combination with jump obstacles, to determine how far the player will jump

"How far" has 2 modes:

- Binary: win/lose. A too small speed will end with the character falling in the pit, but if correct speed is reached, whatever the speed, the character will land exactly at the beginning of the next section of the path
- Progressive: used for 4th level. Here, depending on the speed, the player will land on a different spot, determining their multiplier

Jump obstacles are the representation of this jump moment

They can be basic...

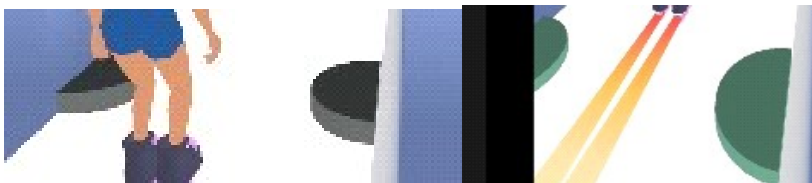


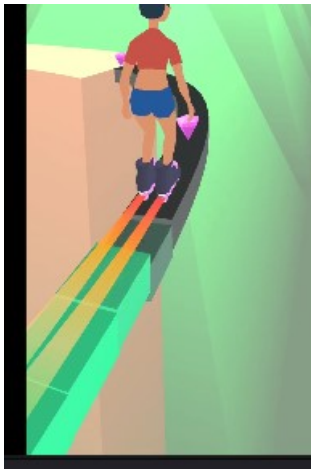
... or associated with a path (normal or progressive)



Visual feedback

To show the player they have properly taken the obstacles, Real obstacles change from black to green. Some speed obstacle are also "visibly activated"





Haptic feedback confirms the visual feedback

Additional “obstacles” and their impact on “perceived difficulty”

Let’s call them Global Obstacles

They are indirect, because their specificity doesn’t lie in taking them or not, but in their influence on the overall situation

Turns

Turns, loops, sloping corners rely on the same effect: reducing the ability for the player to plan, by reducing the scope of the visible path

In my perception, before lvl40 (I don’t know after that), it only affects “perceived difficulty”: it doesn’t really make the level more difficult, but gives the impression of a tension, and also brings variety.

Slope

Slope and sloping corners increase speed, but in a “bad way”: they can’t be avoided, and don’t tell the player “you will jump further” but “watch out, it is going to be more difficult to avoid obstacles”

They can be considered as real sources of difficulty; yet, as the game is quite basic, everywhere we can consider that “difficulty of more speed” = “difficulty of less distance”. Meaning that the same difficulty is added by multiplying the speed by 2 or dividing the distances by 2

Here again, the choice to increase speed instead of reducing distance is to bring more variety.

Meta

Meta can be summarized by actions “unlocking” and “switching”, applied to contents “rollers” and “characters”

Unlock has 2 functions: “enable switching” (for all players) and “collect”, the gotta-catch-them-all effect (for players interested in it)

Switching has only a cosmetic function, either for immediate pleasure (“I like how I look”) or to fight boredom (“I’ve seen this too much”)

Collecting rollers

The means to collect rollers depend on the tab they are in

Except when the contrary is indicated, unowned rollers are “shadow” and owned rollers are “visible”; additionally, a frame is added to the roller currently selected

Random, common and rare

9 items per tab.

With soft-currency, player can launch a random pick

Random pick:

- Cannot land on an already owned item
- Is more expensive for rare than for common
- Is more expensive the $(N+1)^{\text{th}}$ time than the N^{th} time

Chest related: animals

As we said, every 3 chapters, after 3rd level, player enters chestgame

Chestgame is based on 9 chests

Only one of them contain an “additional content”, a roller model or a character model

This potential reward is shown to the player

The other chest contain SC

The player can open 3 chests, corresponding to the 3 keys they have won

With an ad, they can open 3 more

With a second ad, they can open the last 3 ones

Rollers in chest are always animals

Ads based, legendary

Player can choose the item they want, but based on the shadow

To unlock an item, the player must watch 3 ads. Watching 1 or 2, except moving the progress bar, has strictly no effect

Event related

Only one item, and I don't know how it worked

VIP

By purchasing the VIP subscription, the player also has access to 3 additional roller blades

This is the only tab where items are visible even if inaccessible

Collecting characters

Characters work exactly the same way, except that chest related characters are "heroes"

Ads

Ads are the main source of monetization

Banner

Banners are the lowest source of revenue per ad.

There is one banner, on top of the screen, always visible except during video ads

Interstitial

Interstitial video ads appear without the player asking, and for this reason bring less revenue (but more than banners)

They can also be skipped faster

In SR, Interstitials happen at the beginning of a level, including during "restart", after the player has died and refused to see a rewarded video

In my sessions, I have started 51 levels (41 start and 10 restart after death) and experienced 28 interstitials (one every 1.8 level)

Rewarded Videos

Rewarded videos are proposed on specific spots, but launch only if the player asks for it. Player is making a deal with the app: watching a video to get a counterpart. Because the player is supposed to be "in a good state of mind", these videos bring more revenue. They are also longer and longer to skip.

Player can trigger a rewarded video:

- To start from current obstacle, not start, when they die
- To have a multiplied SC reward at the end of the level. The multiplier is given before the video is launched
- To have 3 additional chest keys
- To unlock 1/3 of a special item
- To win 150 SC (can be triggered in common/rare tab)

- *Warning: although data set says there is a random gift rewarded video on some end levels, I never encountered it*

No Ads

“No ads” is a purchase, but as it is related to ads, it is in this section.

“No ads” effect needs to be checked, as it depends on the app (and I didn’t buy it myself in Sky Roller):

- In some apps, it removes interstitial adds
- In some adds, it gives the counterpart of rewarded videos without the need to watch rewarded videos anymore
- In some apps it does both

In Sky Roller, “no ads” can be activated forever through a single purchase (No ads) or be part of a subscription (VIP)

Improvement areas

This section will describe what are, after my short analysis, the main areas for improvement.

It is based on my subjective personal experience playing the 40 first levels, and some comments I found on Google Playstore, in the rating section of the app.

The app is successful, let’s not forget it, so these areas are not “things to do”, but “leads, potential paths” that need to be handled carefully, to avoid harming the game

Most of the current roadmap is based on these improvement areas and on some data analysis.

Soft Currency

There is an issue with Soft-currency.

Soft-currency is supposed to be the main motivation for several aspects of the game:

- Get all gems in levels
- Reach x2 at the end of Gem Rush
- Trigger Rewarded Videos to multiply rewards
- Trigger Rewarded Videos to have additional keys after the special content is discovered
- Trigger Rewarded Videos to win 150 SC in common/rare tab
- Buy VIP status to have 500 SC per day

But it can only be used in one situation: random content tab (normal and rare)

For “collector players”, SC becomes useless as soon as they have all the items

For “cosmetic players”, who wants to only have 1 or 2 item they like, it is even worse: if they unlock these items through ads or VIP program, they will NEVER use soft-currency, creating a huge imbalance between their real motivation to collect SC, and the motivation they are supposed to have.

Addendum, 03/21/22

Carlos has shared some data on Rewarded Videos with me, which seem to contradict part of my analysis. More specifically, triggering Rewarded Videos to win 150 SC in common/rare tab is almost as popular as triggering Rewarded Videos to unlock 1/3 of an item.

Early difficulty curve and learnings

Several small improvements could be done on the early difficulty curve, mainly by distributing more the new elements, or helping the player to learn

Worse situation is level 5:

- This session, in the 40 levels, I lost 10 times. 4 of them were on level 5
- In this session and in the previous install I had done, I lost $3+2=5$ times on the first obs_block_no+full_size2. Nothing has indicated me how to correct it next time. This is even worse because it “looked like” it was good



- Level 5 introduces 5 new obstacles (2-position-blocks x3 & progressive path x2) whereas level 6 and 7 introduce 0 new obstacle
- Progressive path obstacles ask to make 2 progressive moves (full=>no=>full or mid=>full=>mid) but no progressive path obstacle with one move was created and implemented before that (full=>no or no=>full)

Late difficulty curve and content/variety

The opposite happens later on:

- After level 5, I lost only 6 times in 35 levels
- Obstacles are mastered
- New obstacles are not really more difficult
- Variety is low

Main player requests on this subject were to make longer levels, and more varied ones. Make longer level is great because:

- It is statistically more difficult (cumulative odds)
- It gives more “sense” to the rewarded video “don’t restart from beginning”

- It is physically more difficult too: because the tension is longer, the efficiency at the end of the level is lower too, especially if the player has experimented a lot of changes in rhythm and direction. It would by the way make global obstacle more impacting

Characters and rollers

This is not really an issue, but of course, as for every content-driven app, releasing additional content is an improvement for the game, especially if it triggers a new motivation for players that were already happy with what they had